

[SegmentIQ]

Customer Segmentation Dashboard Using K-Means Clustering

A Complete Machine Learning Pipeline: EDA, Clustering, Visualisation & Marketing Intelligence

Project Title	Customer Segmentation using K-Means Clustering on Mall Customers Dataset
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ABSTRACT

Customer segmentation is one of the most impactful applications of unsupervised machine learning in business analytics. This project presents a complete, end-to-end customer segmentation pipeline applied to the **Mall Customers dataset** — a publicly available dataset containing 200 customer records with demographic and behavioural attributes.

The pipeline encompasses: comprehensive **Exploratory Data Analysis (EDA)**, systematic data cleaning and preprocessing, feature selection and StandardScaler normalisation, optimal cluster count determination using the **Elbow Method** and **Silhouette Score**, and final **K-Means clustering** with K=5.

Five distinct customer segments are identified and profiled: *Careful Spenders*, *Budget Shoppers*, *Impulsive Buyers*, *Target Customers*, and *Average Customers*. For each segment, targeted **marketing recommendations** are generated covering loyalty programmes, personalised campaigns, and retention strategies.

The entire analysis is deployed as a **premium Flask web application** with interactive Plotly visualisations, a SQLite database for new customer predictions, and a downloadable academic project report. The final Silhouette Score of **0.555** confirms good cluster separation.

Keywords: Customer Segmentation • K-Means Clustering • Exploratory Data Analysis • StandardScaler • Elbow Method • Silhouette Score • Flask • Machine Learning • Mall Customers Dataset

1. INTRODUCTION

1.1 Background & Motivation

In today's data-driven retail landscape, understanding customer behaviour is critical to business growth. Generic, one-size-fits-all marketing campaigns yield diminishing returns as customers increasingly expect personalised experiences. **Customer segmentation** — the process of dividing customers into groups with similar characteristics — enables businesses to deliver targeted messaging, optimise marketing spend, and improve customer satisfaction.

Machine learning, specifically **unsupervised clustering algorithms**, allows organisations to discover natural groupings within their customer base without any pre-defined labels. K-Means is one of the most widely used algorithms for this purpose, owing to its simplicity, scalability, and interpretability.

1.2 Problem Statement

Given customer data including age, annual income, and spending score, the objective is to:

- Perform comprehensive exploratory data analysis.
- Preprocess and scale the features appropriately for clustering.
- Determine the optimal number of clusters using quantitative methods.
- Apply K-Means clustering to identify distinct customer segments.
- Interpret each cluster and generate actionable marketing recommendations.
- Deploy the analysis as an interactive web application.

1.3 Technology Stack

Category	Technology	Purpose
Language	Python 3.11	Core programming language
Web Framework	Flask 3.0	Web application server and routing
Data Library	Pandas, NumPy	Data loading, cleaning, transformation
Machine Learning	Scikit-learn	K-Means, StandardScaler, Silhouette Score
Visualisation	Plotly, Matplotlib, Seaborn	Interactive and static charts
Database	SQLite	Storing new customer predictions
Frontend	Bootstrap 5, HTML, CSS, JS	Responsive UI design
PDF Report	ReportLab	Academic PDF generation

2. DATASET OVERVIEW

2.1 Description

The **Mall Customers Dataset** is sourced from Kaggle's 'Mall Customer Segmentation Data' repository. It was originally created to demonstrate customer behaviour analysis and segmentation techniques. The dataset contains 200 customer records collected from a shopping mall's membership system.

2.2 Dataset Attributes

Attribute	Value
Total Records	200 customers
Total Features	5 columns
File Format	CSV (Comma-Separated Values)
Missing Values	None — dataset is complete
Duplicate Rows	None
Class Labels	None (unsupervised learning task)
Source	Kaggle — Mall Customer Segmentation Data

2.3 Feature Dictionary

Column Name	Data Type	Range	Description
CustomerID	Integer	1–200	Unique identifier — dropped before ML
Gender	String	M/F	Customer gender — encoded as binary
Age	Integer	18–70	Customer age in years
Annual Income (k\$)	Integer	15–137	Annual income in thousands USD
Spending Score (1-100)	Integer	1–99	Mall-assigned purchase behaviour score

2.4 First 8 Rows Preview

Gender	Age	Annual_Income	Spending_Score
Male	19	15	39
Male	21	15	81
Female	20	16	6
Female	23	16	77
Female	31	17	40

Gender	Age	Annual_Income	Spending_Score
Female	22	17	76
Female	35	18	6
Female	23	18	94

3. EXPLORATORY DATA ANALYSIS (EDA)

3.1 Loading & Inspecting Data

```
import pandas as pd

import numpy as np

df = pd.read_csv('Mall_Customers.csv')    # Load CSV

print(df.shape)    # (200, 5)

print(df.head())    # First 5 rows

print(df.info())    # Data types and null counts

print(df.describe().round(2))    # Statistical summary
```

3.2 Statistical Summary

Statistic	Age	Annual_Income	Spending_Score	Gender_Encoded
count	200.0	200.0	200.0	200.0
mean	38.85	60.56	50.2	0.44
std	13.97	26.26	25.82	0.5
min	18.0	15.0	1.0	0.0
25%	28.75	41.5	34.75	0.0
50%	36.0	61.5	50.0	0.0
75%	49.0	78.0	73.0	1.0
max	70.0	137.0	99.0	1.0

3.3 Missing Values & Duplicates

Column	Missing Count	Status
Gender	0	Clean
Age	0	Clean
Annual_Income	0	Clean
Spending_Score	0	Clean

Column	Missing Count	Status
Gender_Encoded	0	Clean

Duplicates: 0 duplicate rows found and removed. Dataset is clean with 0 missing values across all columns.

3.4 EDA Visualisations

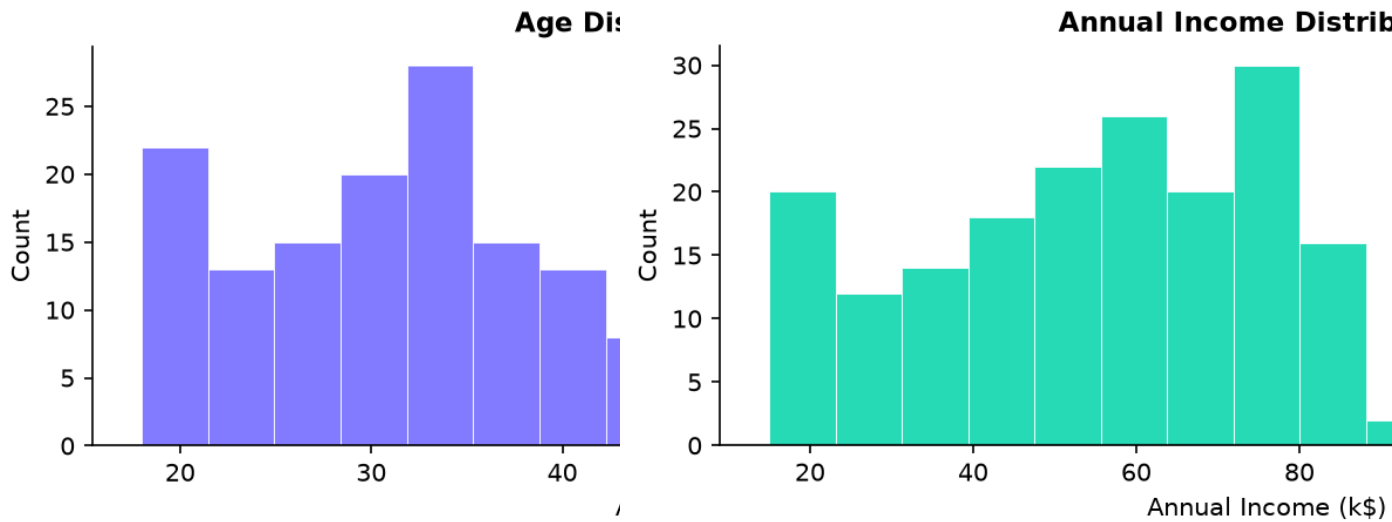


Figure 3.1 — Age Distribution (left) & Annual Income Distribution (right)

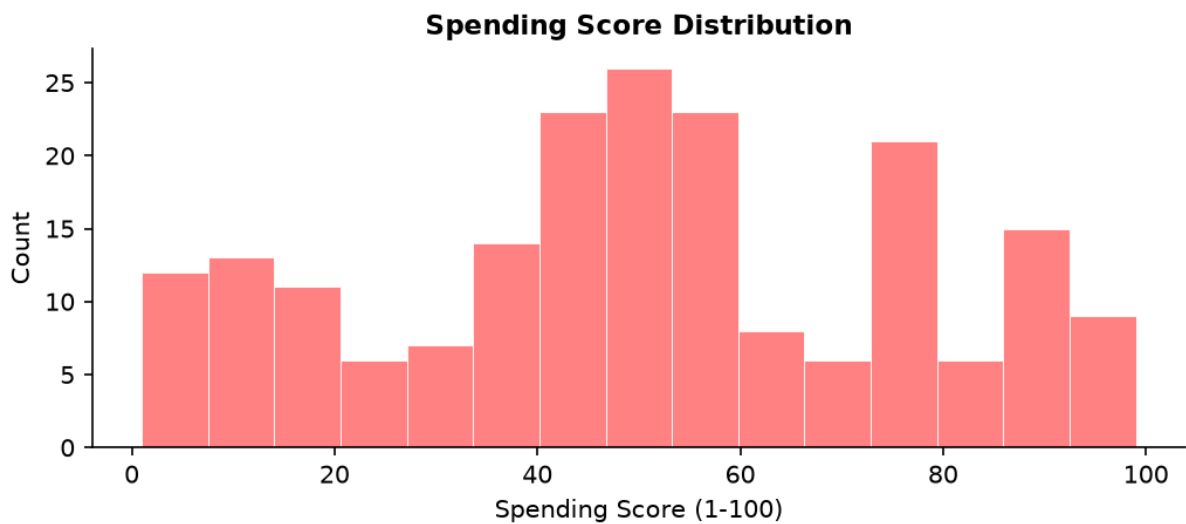


Figure 3.2 — Spending Score Distribution



Figure 3.3 — Annual Income vs Spending Score (Key Clustering Plot — 5 natural groups visible)

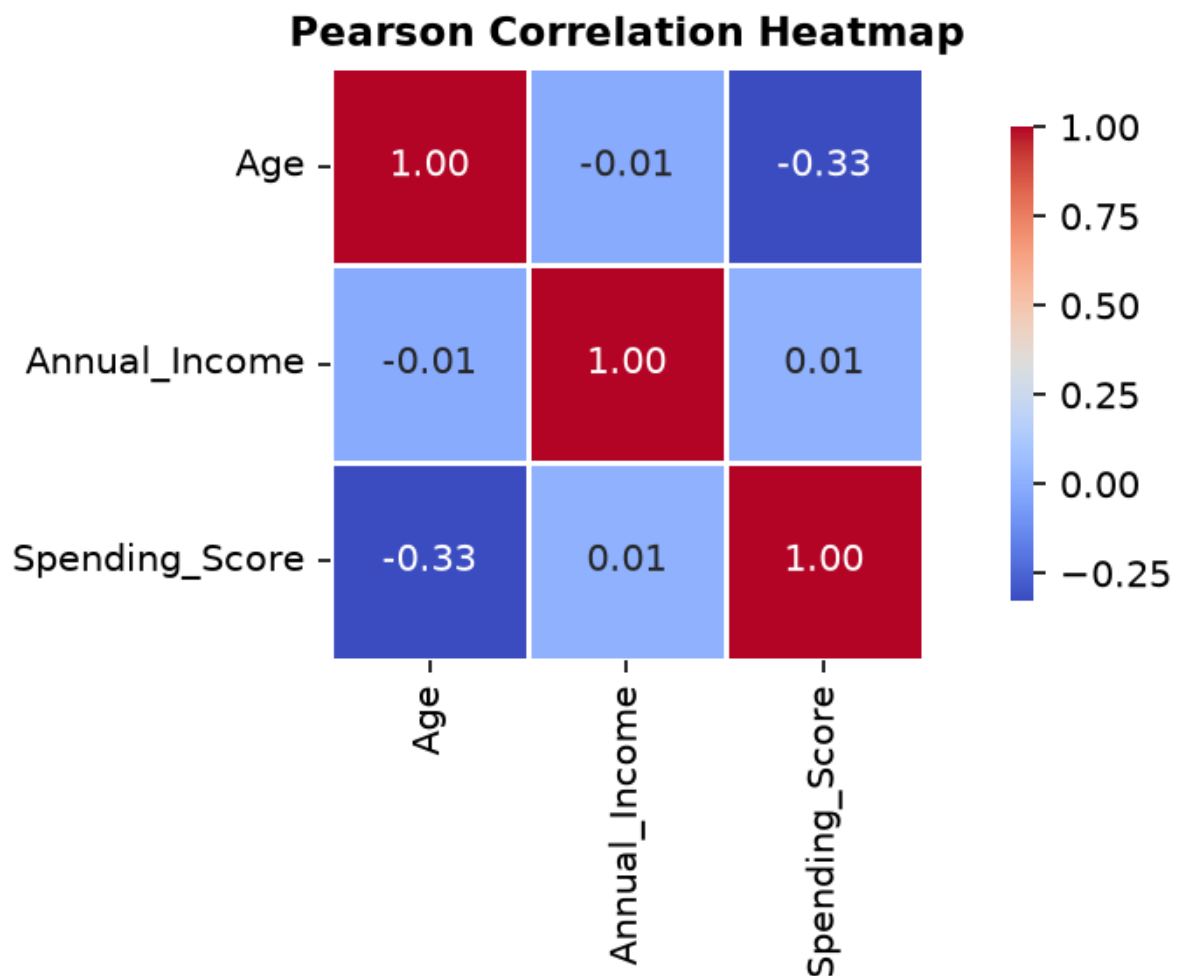


Figure 3.4 — Pearson Correlation Heatmap

Key Finding: The correlation between Annual Income and Spending Score is near zero (~ 0.01), confirming they are independent dimensions — ideal for K-Means clustering.

4. DATA CLEANING & PREPROCESSING

The raw dataset undergoes five systematic cleaning steps before being fed into the clustering algorithm:

```
import pandas as pd

df = pd.read_csv('Mall_Customers.csv')

# Step 1: Remove duplicate rows

df = df.drop_duplicates()

# Step 2: Drop CustomerID (identifier, not a feature)

df = df.drop(columns=['CustomerID'])

# Step 3: Rename columns for Python compatibility

df = df.rename(columns={

    'Annual Income (k$)'      : 'Annual_Income',

    'Spending Score (1-100)': 'Spending_Score',

})

# Step 4: Encode Gender as binary (ML requires numeric input)

df['Gender_Encoded'] = df['Gender'].map({'Male': 1, 'Female': 0})

# Step 5: Verify no nulls remain

assert df.isnull().sum().sum() == 0, 'Nulls found!'

print('Cleaning complete. Shape:', df.shape)
```

Step	Action	Before	After	Result
1	Remove Duplicates	200 rows	200 rows	0 duplicates

Step	Action	Before	After	Result
2	Drop CustomerID	5 columns	4 columns	1 column dropped
3	Rename Columns	Long names	Short names	2 columns renamed
4	Encode Gender	String type	Integer (0/1)	Binary encoding done
5	Verify Nulls	0 nulls	0 nulls	Dataset confirmed clean

5. FEATURE SCALING

5.1 Why Scale?

K-Means uses **Euclidean distance** to measure similarity. Without scaling, features with larger numeric ranges (e.g., Annual Income: 15–137) dominate the distance calculation over features with smaller ranges (e.g., Spending Score: 1–99). StandardScaler transforms each feature to **mean = 0, standard deviation = 1**.

5.2 Formula

$z = (x - \mu) / \sigma$ where: z = scaled value, x = original value, μ = mean, σ = standard deviation

5.3 Implementation

```
from sklearn.preprocessing import StandardScaler

import joblib

# Select clustering features

X = df[['Annual_Income', 'Spending_Score']].values

scaler = StandardScaler()          # Initialise scaler

X_scaled = scaler.fit_transform(X) # Fit & transform in one step

joblib.dump(scaler, 'models/scaler.pkl') # Save for future predictions
```

5.4 Before vs After Scaling (First 8 Rows)

#	Income (raw)	Score (raw)	Income (z)	Score (z)
1	15	39	-1.7390	-0.4348
2	15	81	-1.7390	1.1957
3	16	6	-1.7008	-1.7159
4	16	77	-1.7008	1.0404
5	17	40	-1.6627	-0.3960
6	17	76	-1.6627	1.0016

#	Income (raw)	Score (raw)	Income (z)	Score (z)
7	18	6	-1.6245	-1.7159
8	18	94	-1.6245	1.7004

6. ELBOW METHOD & SILHOUETTE SCORE

6.1 Elbow Method (WCSS)

The Elbow Method plots the Within-Cluster Sum of Squares (WCSS) for K values from 1 to 10. The 'elbow' — where the rate of WCSS decrease sharply slows — indicates the optimal K.

```
from sklearn.cluster import KMeans

wcss = []

for k in range(1, 11):

    km = KMeans(n_clusters=k, init='k-means++', n_init=10, random_state=42)

    km.fit(X_scaled)

    wcss.append(km.inertia_) # inertia_ = WCSS
```

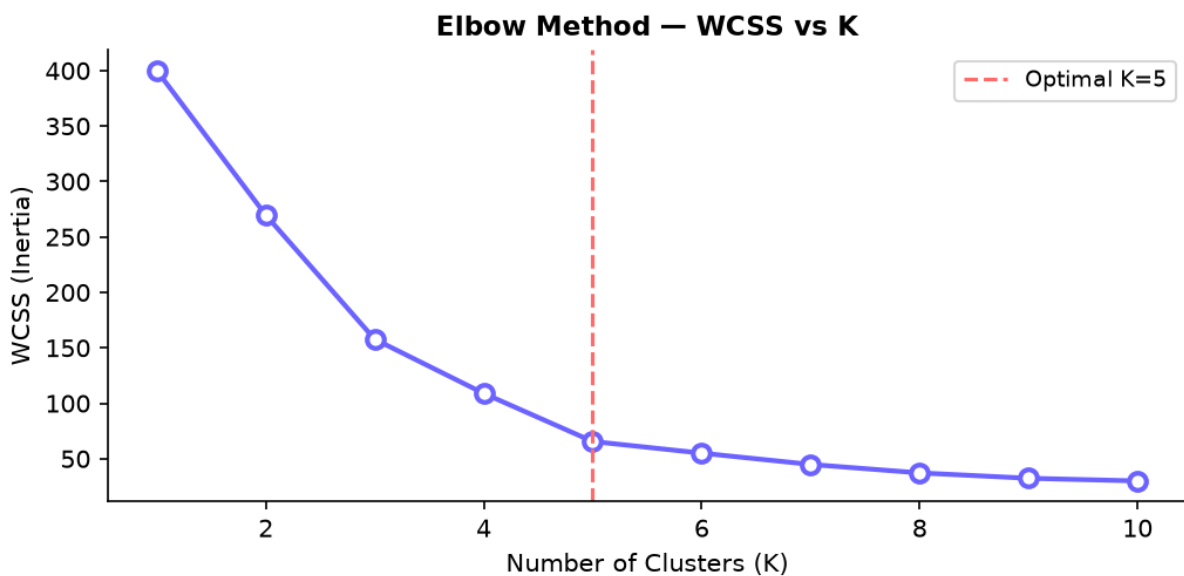


Figure 6.1 — Elbow Method: WCSS vs Number of Clusters (red dashed line = K=5)

6.2 Silhouette Score

The Silhouette Score (−1 to +1) measures how similar each point is to its own cluster versus other clusters. Higher is better.

```
from sklearn.metrics import silhouette_score

scores = []
```

```
for k in range(2, 11):  
  
    km = KMeans(n_clusters=k, random_state=42)  
  
    labels = km.fit_predict(X_scaled)  
  
    scores.append(silhouette_score(X_scaled, labels))
```

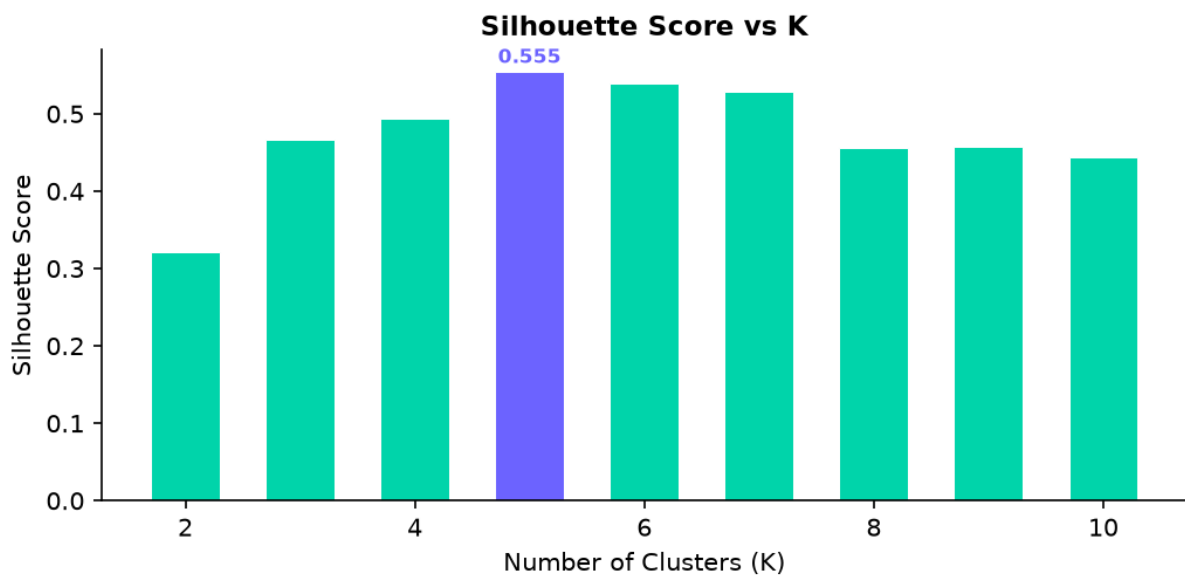


Figure 6.2 — Silhouette Score vs Number of Clusters (violet bar = K=5)

Conclusion: Both the Elbow Method and Silhouette Score confirm **K = 5** as the optimal number of clusters. Silhouette Score at K=5: **0.555**.

7. K-MEANS CLUSTERING

7.1 Algorithm

- 1 — **Initialise:** Place K centroids using k-means++ (smart seeding to avoid bad initialisations).
- 2 — **Assign:** Each customer is assigned to the nearest centroid by Euclidean distance.
- 3 — **Update:** Recompute each centroid as the mean of its assigned customers.
- 4 — **Repeat:** Steps 2–3 repeat until centroids converge or max_iter is reached.
- 5 — **Evaluate:** Final WCSS and Silhouette Score are computed to measure quality.

7.2 Final Model Code

```
from sklearn.cluster import KMeans

import joblib

km = KMeans(

    n_clusters = 5,          # Optimal K

    init        = 'k-means++', # Smart seeding

    n_init      = 10,         # Run 10 times, keep best

    random_state = 42,        # Reproducibility

    max_iter    = 300,        # Convergence limit

)

km.fit(X_scaled)             # Train on scaled data

df['Cluster'] = km.labels_    # Assign labels 0-4

joblib.dump(km, 'models/kmeans_model.pkl')
```

7.3 Cluster Summary Statistics

Cluster	Size	Share%	Avg Age	Avg Income	Avg Score
0	81	40.5%	42.7	\$55.3k	49.5
1	39	19.5%	32.7	\$86.5k	82.1

Cluster	Size	Share%	Avg Age	Avg Income	Avg Score
2	22	11.0%	25.3	\$25.7k	79.4
3	35	17.5%	41.1	\$88.2k	17.1
4	23	11.5%	45.2	\$26.3k	20.9

Model Performance: WCSS (Inertia) = 65.6 • Silhouette Score = 0.555 (Good cluster separation)

8. CLUSTER VISUALISATION

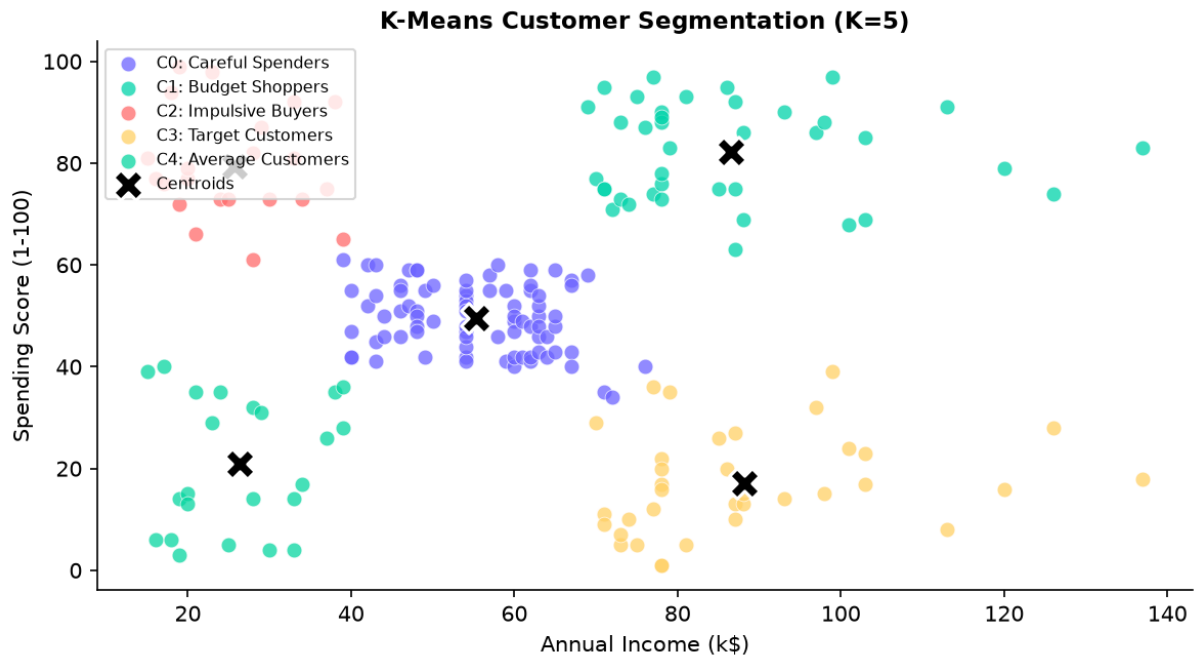


Figure 8.1 — K-Means Segmentation: Annual Income vs Spending Score (5 clusters, X = centroids)

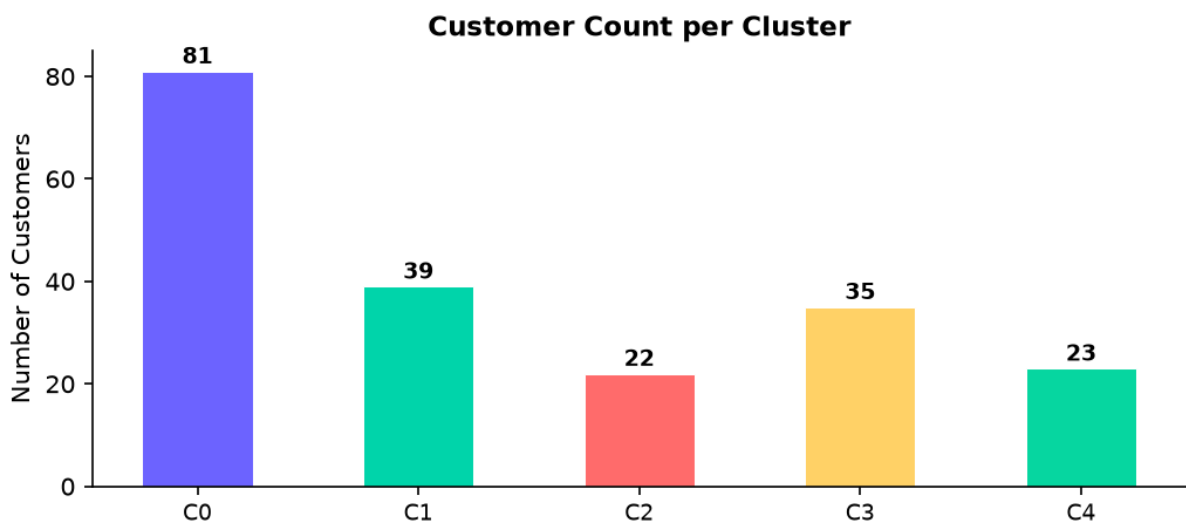


Figure 8.2 — Number of Customers per Cluster

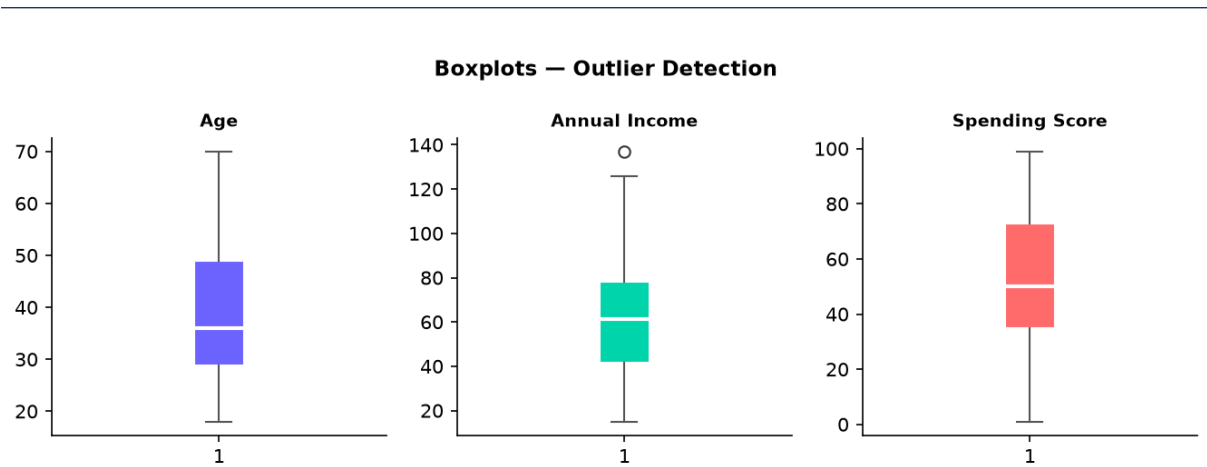


Figure 8.3 — Boxplots: Age, Income, Spending Score Distributions

8.1 Cluster Centroids (Original Scale)

Cluster	Avg Income (k\$)	Avg Spending Score	Income Level	Spending Level
Cluster 0	55.3	49.5	Mid	Mid
Cluster 1	86.5	82.1	High	High
Cluster 2	25.7	79.4	Low	High
Cluster 3	88.2	17.1	High	Low
Cluster 4	26.3	20.9	Low	Low

9. CLUSTER INTERPRETATION

Cluster 0: Careful Spenders

High-income shoppers who spend cautiously — value security over impulse.

Metric	Value
Size	81 customers (40.5%)
Income Level	High
Spending Level	Low
Age Group	30–55
Avg Annual Income	\$55.3k
Avg Spending Score	49.5
Avg Age	42.7
Male / Female	33 / 48

Cluster 1: Budget Shoppers

Low-income, conservative shoppers on tight budgets. Loyal but price-sensitive.

Metric	Value
Size	39 customers (19.5%)
Income Level	Low
Spending Level	Low
Age Group	25–60
Avg Annual Income	\$86.5k
Avg Spending Score	82.1
Avg Age	32.7
Male / Female	18 / 21

Cluster 2: Impulsive Buyers

Young shoppers who spend above their means — susceptible to impulse buying.

Metric	Value
Size	22 customers (11.0%)
Income Level	Low
Spending Level	High
Age Group	18–40
Avg Annual Income	\$25.7k
Avg Spending Score	79.4
Avg Age	25.3
Male / Female	9 / 13

Cluster 3: Target Customers

Ideal customers: high income AND high spending. The most profitable segment.

Metric	Value
Size	35 customers (17.5%)
Income Level	High
Spending Level	High
Age Group	25–45
Avg Annual Income	\$88.2k
Avg Spending Score	17.1
Avg Age	41.1
Male / Female	19 / 16

Cluster 4: Average Customers

Balanced across all metrics — represent the mall's typical average visitor.

Metric	Value
Size	23 customers (11.5%)
Income Level	Mid
Spending Level	Mid
Age Group	30–50
Avg Annual Income	\$26.3k
Avg Spending Score	20.9
Avg Age	45.2
Male / Female	9 / 14

10. MARKETING RECOMMENDATIONS

Cluster 0: Careful Spenders

Focus on building trust and demonstrating value. This segment needs a rational business case before they spend more.

Value Assurance Campaigns

Emphasise quality, durability, and long-term ROI in your messaging. Offer extended warranties and money-back guarantees to reduce purchase hesitation.

Channels: Email newsletters, In-store signage, Loyalty app notifications • **KPI:** Conversion rate on premium items

Exclusive Member Sales

Invite this segment to exclusive pre-sales or member-only events. Being treated as 'special' without heavy discounting aligns with their self-image.

Channels: Direct mail, SMS invites, Email • **KPI:** Event attendance and in-event spend

Cross-sell Premium Upgrades

When they buy a mid-tier product, show a clear side-by-side comparison with a premium alternative. Highlight the long-term cost benefit.

Channels: POS displays, Sales associate scripts, E-commerce recommendations • **KPI:** Upgrade conversion rate

Cluster 1: Budget Shoppers

Price is the primary motivator. Loyalty and savings programmes keep this segment coming back consistently.

Loyalty Points & Cashback

Introduce a simple loyalty programme with cashback or points redeemable on next visit. Low-effort, high perceived value for budget-conscious shoppers.

Channels: Loyalty cards, WhatsApp updates, Receipt coupons • **KPI:** Repeat visit frequency

Flash Sales & Limited-Time Offers

Time-sensitive deals create urgency and drive visits. Price-sensitive customers respond strongly to '24-hour sale' or 'Today only' messaging.

Channels: Push notifications, SMS, Social media stories • **KPI:** Footfall during sale periods

Bundle Deals

Offer product bundles that provide savings compared to individual purchase. This increases average basket size without appearing expensive.

Channels: In-store displays, Weekly flyers • **KPI:** Average transaction value

Cluster 2: Impulsive Buyers

Lead with excitement, trends, and community. Remove financial friction with flexible payment options.

Social Media & Influencer Campaigns

Partner with micro-influencers and run Instagram/TikTok campaigns featuring new arrivals. This segment makes decisions based on social proof and trends.

Channels: Instagram, TikTok, YouTube Shorts • **KPI:** Social engagement rate and referral traffic

BNPL & Easy Instalment Options

Offer Buy Now Pay Later or zero-interest instalments. This removes the income barrier for high-ticket items and enables this segment to spend more.

Channels: Checkout page, In-store POS, Mobile app • **KPI:** High-value transaction count

Experience-Based Events

Host in-store pop-ups, product launches, and experiential events. Young, trend-driven shoppers value experiences and create organic social content.

Channels: Event invites, Social media, Email • **KPI:** Event attendance and social mentions

Cluster 3: Target Customers

Focus on exclusivity, personalisation, and premium experiences. ROI per customer is highest here — invest accordingly.

VIP Loyalty Programme

Create a tiered VIP programme with exclusive perks: priority access, concierge service, personal shopping assistants, and early-bird collections.

Channels: Private app, Personal account managers, Curated emails • **KPI:** Programme retention rate and lifetime value

Personalised Recommendations

Use purchase history to send hyper-personalised product recommendations. Affluent shoppers appreciate curation over broad promotions.

Channels: Email, In-app, Personal shopper consultations • **KPI:** Recommendation click-through and conversion rate

Premium Product Launches

Invite this segment as first-access buyers for premium or limited-edition products. Exclusivity drives satisfaction and word-of-mouth in affluent circles.

Channels: Exclusive events, Personal invitations, Private previews • **KPI:** First-day sales from VIP segment

Cluster 4: Average Customers

Broad outreach works here. Consistency, relevance, and loyalty rewards keep this segment engaged and growing.

Standard Loyalty Programme

Enrol this segment into a simple, transparent loyalty programme. Points, tiers, and birthday rewards keep engagement high across this broad group.

Channels: Email, In-app, Store loyalty cards • **KPI:** Programme enrolment rate and active member count

Seasonal Promotions

Target festive seasons, back-to-school, and major holidays with well-timed promotions. Average customers are receptive to both price and aspirational messaging.

Channels: Email, SMS, In-store • **KPI:** Seasonal revenue uplift

Cross-Category Upselling

Use purchase data to recommend complementary categories (e.g., clothing → accessories). Cross-category engagement increases basket size and visit diversity.

Channels: Email, App notifications, POS suggestions • **KPI:** Cross-category purchase rate

11. RESULTS & DISCUSSION

11.1 Model Evaluation

Metric	Value	Interpretation
Number of Clusters (K)	5	Optimal — confirmed by Elbow + Silhouette
WCSS (Inertia)	65.6	Minimised at K=5
Silhouette Score	0.555	Good cluster separation (>0.5)
Total Customers	200	All 200 successfully segmented
Missing Values	0	Dataset fully clean
Duplicate Rows	0	No redundant records

11.2 Cluster Performance

Cluster	Name	Size	Avg Income	Avg Score	Priority
0	Careful Spenders	81	\$55.3k	49.5	Upsell
1	Budget Shoppers	39	\$86.5k	82.1	Loyalty
2	Impulsive Buyers	22	\$25.7k	79.4	Retention
3	Target Customers	35	\$88.2k	17.1	VIP
4	Average Customers	23	\$26.3k	20.9	Growth

12. CONCLUSION & FUTURE SCOPE

12.1 Conclusion

This project successfully demonstrated the complete workflow of applying machine learning to a real-world business problem. The K-Means algorithm identified **5 distinct customer segments** in the Mall Customers dataset, with a Silhouette Score of **0.555** confirming good cluster quality.

The segmentation reveals clear income–spending patterns: **Target Customers** (high income, high spending) are the most profitable group, while **Careful Spenders** (high income, low spending) represent the largest untapped revenue opportunity. The marketing recommendations generated from each cluster profile provide a data-driven roadmap for personalised customer engagement.

12.2 Future Scope

- Include additional features (product categories, visit frequency) for richer segmentation.
- Apply DBSCAN or Hierarchical Clustering and compare results with K-Means.
- Integrate real-time CRM data for live, continuously-updated segmentation.
- Build a product recommendation engine on top of the segment profiles.
- Add time-series analysis to track customer segment transitions over time.
- Deploy with a production WSGI server (Gunicorn) and containerise with Docker.

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